

Fast manipulation of Bose-Einstein condensates

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UNDER THE SUPERVISION OF ROBIN CORGIER

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Plan

Castin-Dum equations

reverse-engineering and Shortcut To Adiabaticity method

Painted potentials

Experimental limitations

Future Objectives

Castin-Dum equations

$$\text{Gross-Pitaevskii : } \mu\phi(\mathbf{r}) = \left(-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{ext}}(\mathbf{r}) + Ng|\phi(\mathbf{r})|^2 \right) \phi(\mathbf{r})$$

$$\text{Gross-Pitaevskii (Thomas-Fermi) : } \phi_{TF}(\mathbf{r}, 0) = \sqrt{\frac{\mu - V_{\text{ext}}(\mathbf{r}, 0)}{Ng}}$$

$$\text{Thomas-Fermi radius : } R_{TF}^j(0) = \sqrt{\frac{2\mu}{m\omega_j^2(0)}}$$

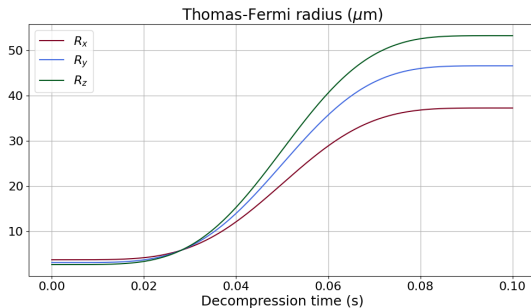
$$\text{Scaling law : } R_{TF}^j(t) = \lambda_j(t)R_{TF}^j(0)$$

$$\text{Castin-Dum : } \ddot{\lambda}_j(t) + \omega_j^2(t)\lambda_j(t) = \frac{\omega_j^2(0)}{\lambda_x(t)\lambda_y(t)\lambda_z(t)\lambda_j(t)}$$

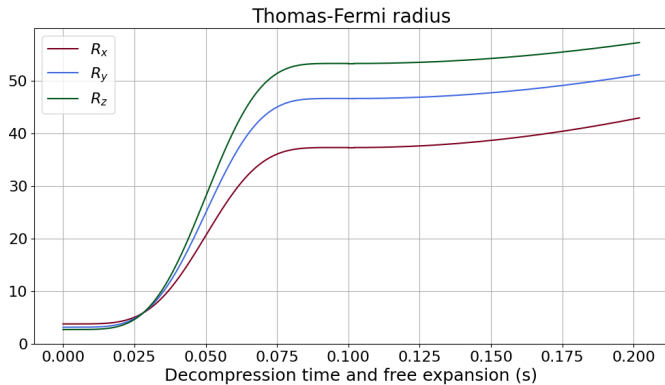
Reverse-engineering

$$\lambda(t) = \lambda(0) + \frac{\lambda(t_f) - \lambda(0)}{12\pi} (6v - 8\sin(v) + \sin(2v))$$

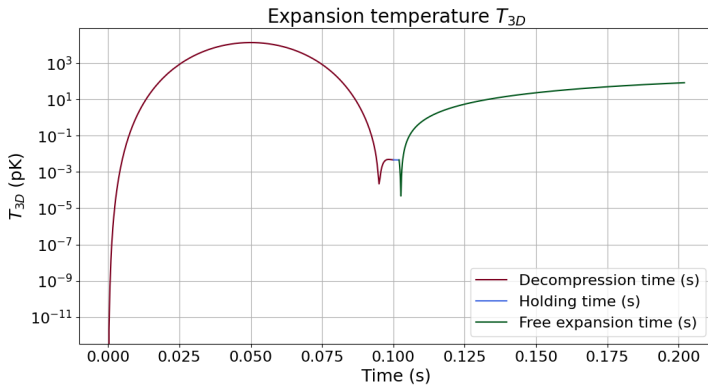
$$v(t) = 2\pi \frac{1 + a\frac{t}{t_f} + b\frac{t^2}{t_f^2}}{1 + a + b} \frac{t}{t_f}$$



Reverse-engineering



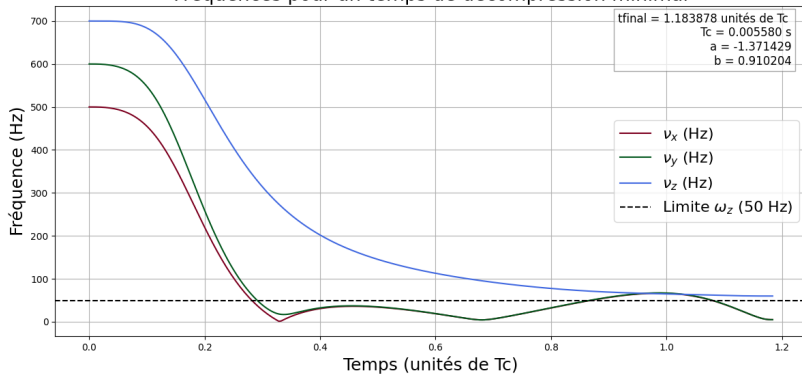
Reverse-engineering



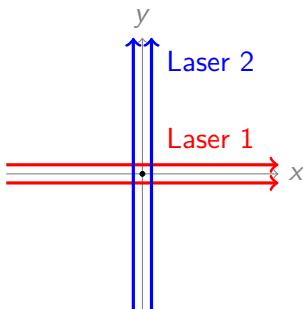
Shortcut to adiabaticity

$$T_c = \frac{1}{4\sqrt{2}} \max \left\{ \left| \frac{1}{\omega_j(t_f)} - \frac{1}{\omega_j(0)} \right| \right\}$$

Fréquences pour un temps de décompression minimal



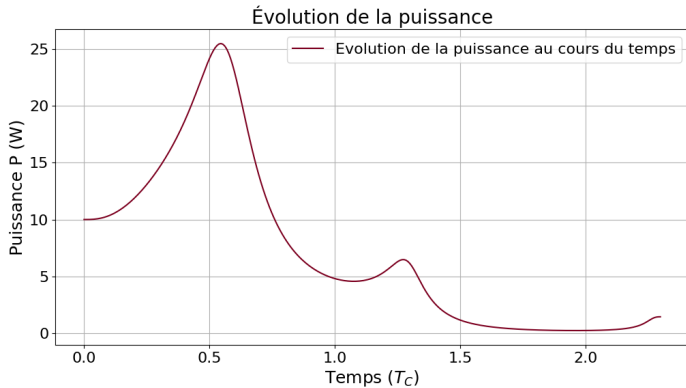
Painted potentials



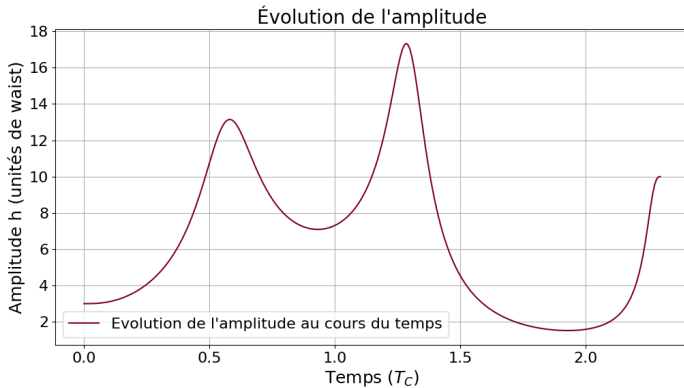
$$\omega_H^2(t) = -\frac{3\sqrt{\pi}U_0P(t)}{m\pi w_V\sqrt{2}h^3(t)}$$

$$\omega_V^2(t) = -\frac{12\sqrt{\pi}U_0P(t)}{m\pi w_V^3\sqrt{2}h(t)}$$

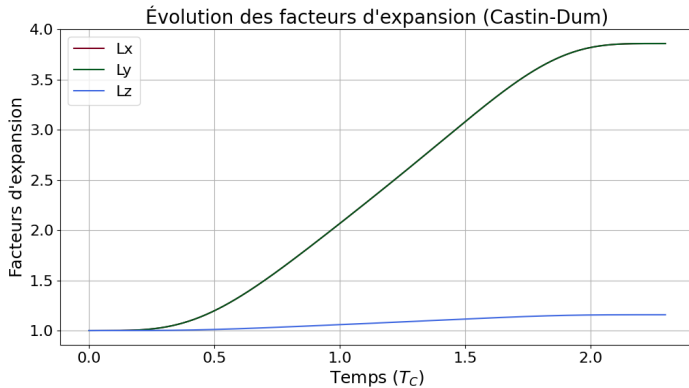
Painted potentials



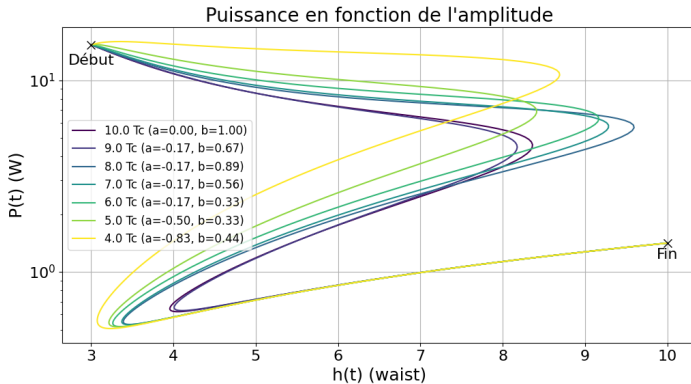
Painted potentials



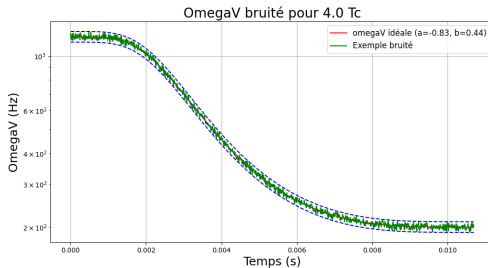
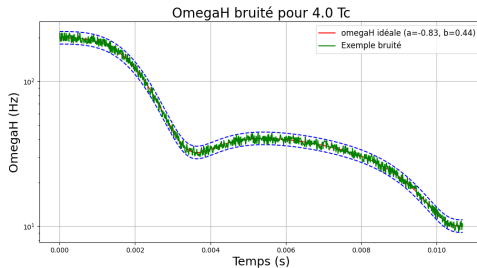
Painted potentials



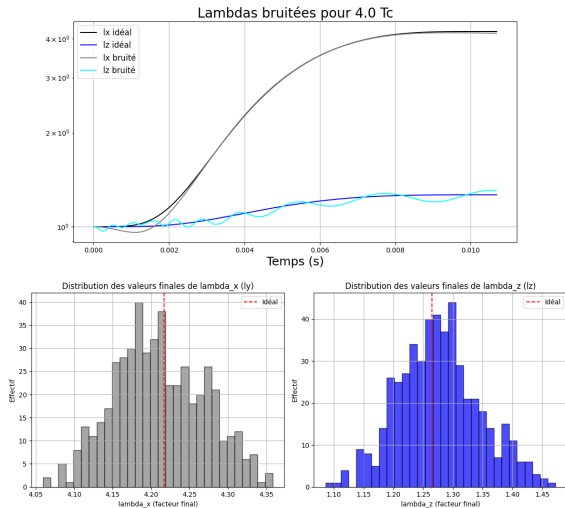
Painted potentials



Experimental limitations



Experimental limitations



Future Objectives

Quantify laser misalignment effects.

Numerically solve the painting problem ($< 3 \times w_V$ limit).

Find the best inverse engineering solution.